SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO3_AT1 — EXPERIMENT

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO3 – Construction (BL3)	**Assessment:**	Assessment Tool 1 – Experiment
Weightage:	20% of CIA	**Total Marks:**	25 (5 Questions × 5 Marks)
CO3 Statement:	*Apply appropriate experimental charting techniques — pie charts, bean plots, violin plots, box plots, and stacked bar charts — to construct accurate visualizations from given data.(BL3)*		
Student Name:	Nivya Sri T	**Reg. No.:**	192424131
Section / Batch:	B.Tech AIDS	**Date:**	11/08/26

Code of Conduct

I Nivya Sri T (192424131) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____Nivya Sri T____

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 75 minutes.
- Graph paper or a ruler is recommended for accurate, to-scale drawing.
- Show all calculations (e.g., pie chart angles, five-number summary, IQR) as part of your answer.
- Every chart must include a title, correctly labeled axes (where applicable), and a legend where relevant.

Annexure A – EXPERIMENT

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT1 — Experiment

CO3 · BT3: Apply ·

This rubric is used to assess student responses to the CO3-AT1 Experiment, in which students are given a small dataset and must construct an accurate pie chart, bean plot, violin plot, box plot, or stacked bar chart.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25%**	Demonstrates thorough, accurate understanding of the correct chart type (pie, bean, violin, box, or stacked bar) and the elements required to construct it for the given data.	Good understanding of the required chart type and construction elements, with only minor gaps.	Basic understanding of chart construction, but with some misconceptions about the correct chart type or required elements.	Poor understanding of core construction concepts; selects a chart type fundamentally unsuited to the data.
Explanation	**25%**	Shows detailed, clearly worked calculations (e.g., pie chart angles, five-number summary, IQR, stacked totals) and explains each construction step.	Shows clear calculations and construction steps, with adequate explanation for most parts.	Calculations or construction steps are shown but incomplete, or explanations are superficial.	Little or no working shown; calculations and construction steps are missing or unexplained.
Application	**20%**	Effectively applies chart construction techniques to produce a complete, accurate chart correctly matched to the data.	Applies construction techniques to produce a correct chart, with only minor errors.	Limited application of construction techniques; the chart partially represents the data but has notable errors.	Fails to apply appropriate construction techniques; the chart does not correctly represent the data.
Organization	**15%**	The chart is neatly and logically constructed, with a title, correctly labeled axes or slices, a legend where applicable, and elements in a clear order.	The chart is well organized, with only minor issues in labeling, legend, or layout.	Basic construction; required elements (title, labels, legend) are present but poorly arranged or partly missing.	Poorly constructed; the chart is missing key elements such as a title, labels, or a legend.
Accuracy	**10%**	The chart is completely accurate — all values, proportions, quartiles, and calculations	Mostly accurate, with only minor omissions or a small error in one value or calculation.	Partially correct; some plotted values, proportions, or calculations are missing or incorrect.	Largely inaccurate; major errors in plotted values, proportions, or calculations relative to the given data.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		correctly match the given data.			

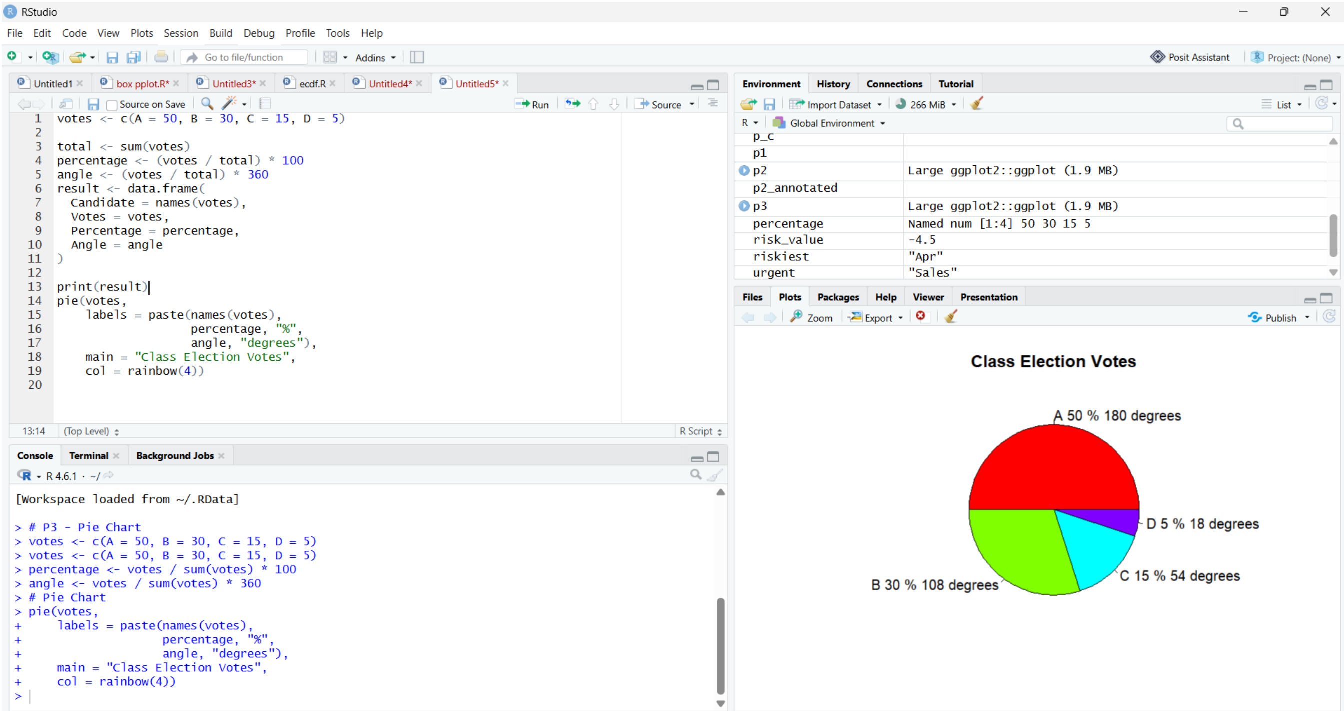
Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _______________	Signature: _______________	Signature: _______________
Date: ___________________	Date: ___________________	Date: ___________________

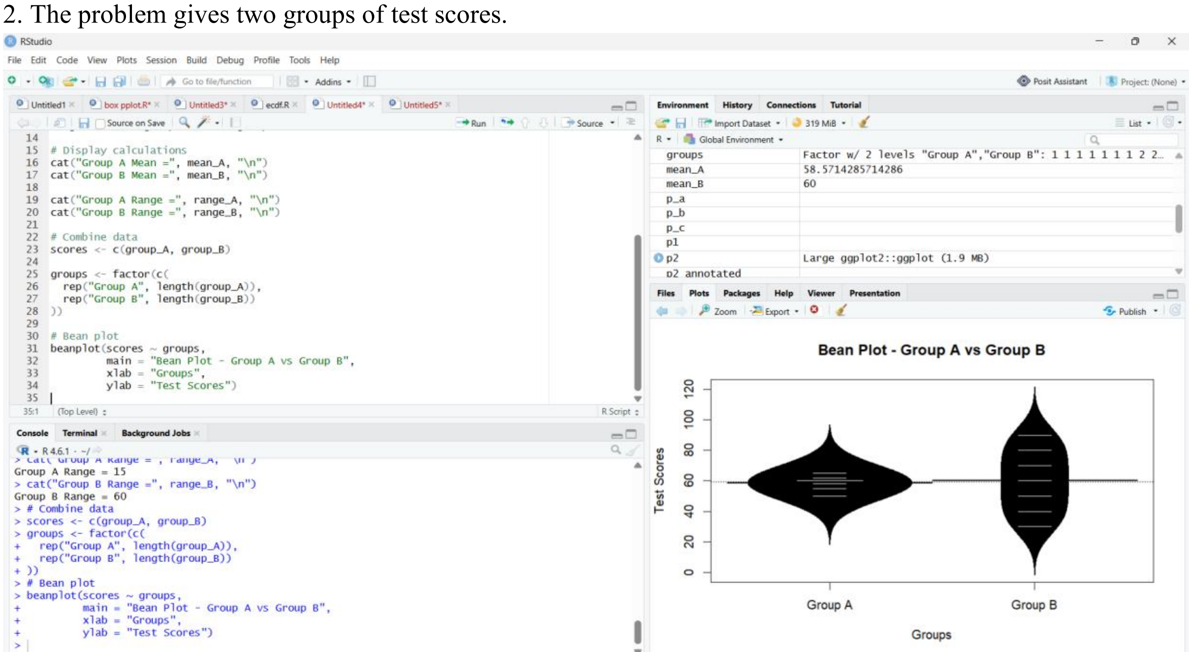
1.

The problem gives votes A = 50, B = 30, C = 15, D = 5.



Candidate	**Votes**	**Percentage**	**Angle**
A	50	50%	180°
B	30	30%	108°
C	15	15%	54°
D	5	5%	18°
Total	**100**	**100%**	**360°**

2. The problem gives two groups of test scores.

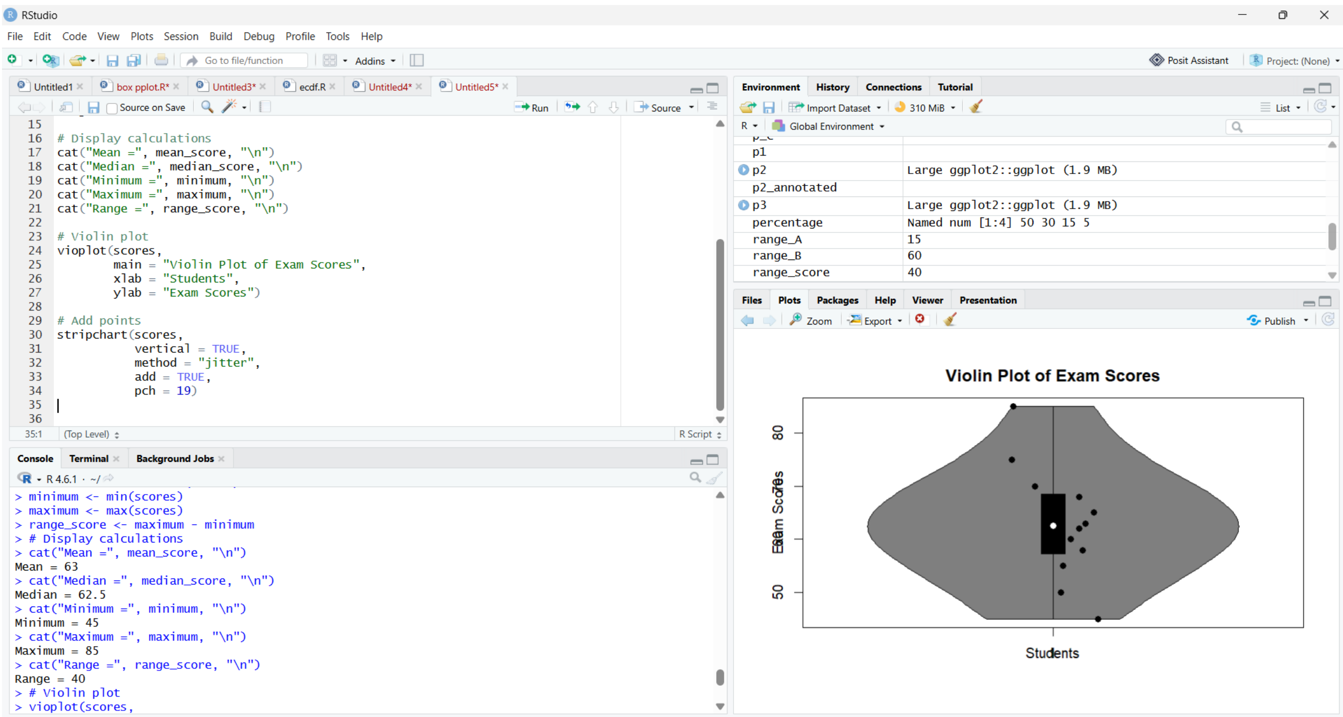


Group A Mean = 58.57143
Group B Mean = 60

Group A Range = 15
Group B Range = 60

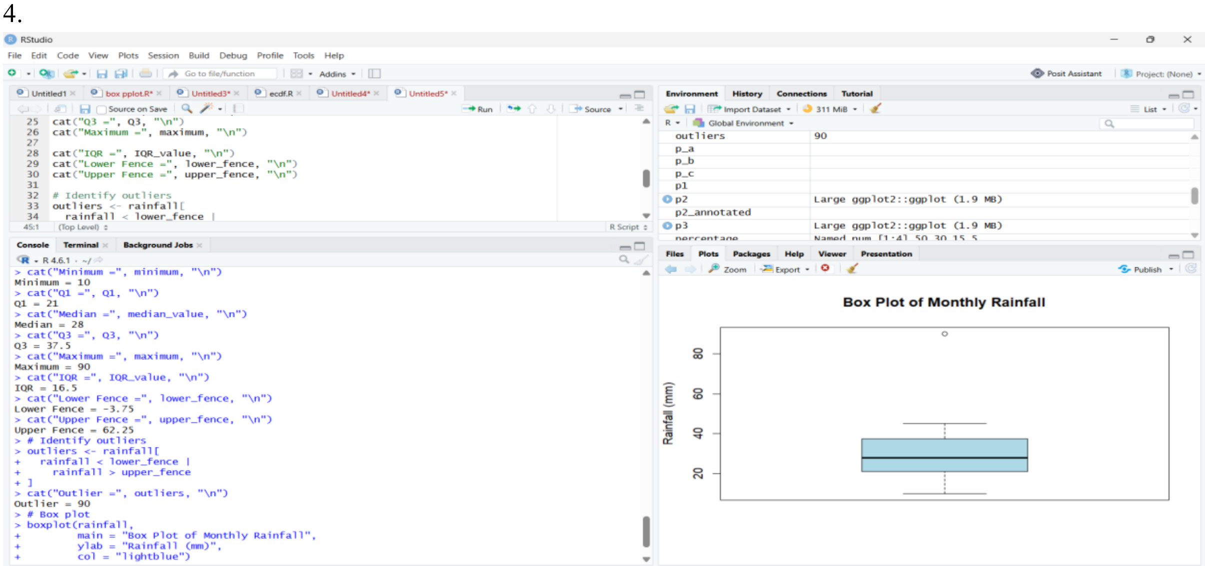
Observation: The means are almost similar, but Group B has much greater spread.

3.



Mean = 63
Median = 62.5
Minimum = 45
Maximum = 85
Range = 40

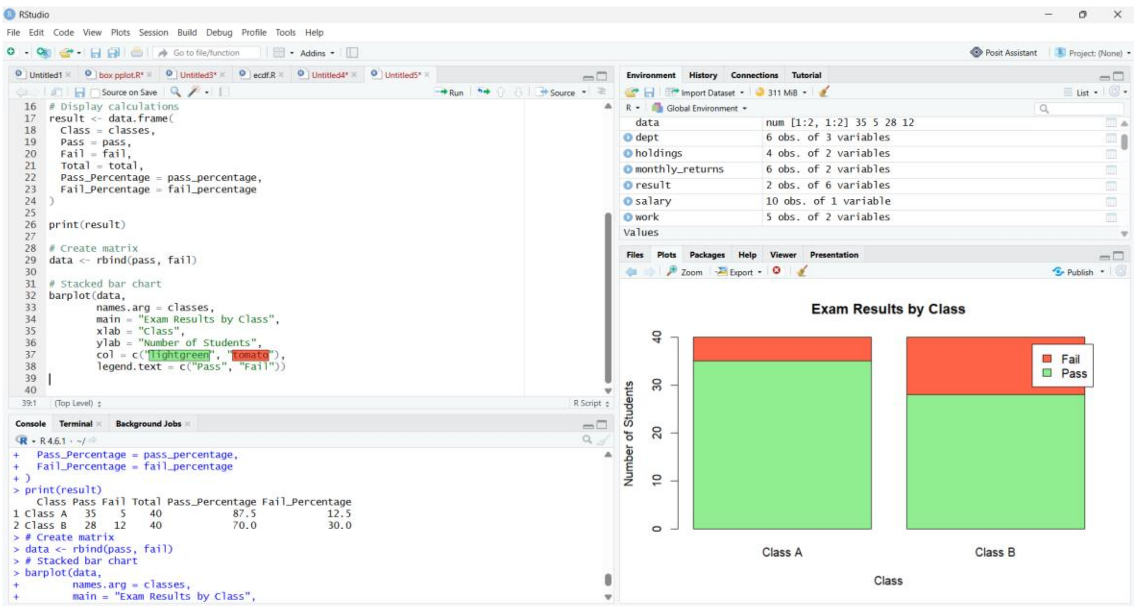
4.



Minimum = 10
Q1 = 21
Median = 28
Q3 = 37.5
Maximum = 90
IQR = 16.5
Lower Fence = -3.75
Upper Fence = 62.25

Outlier = 90
So **90 mm is the outlier**.

5.



Class	Pass	Fail	Total	Pass Percentage	Fail Percentage
Class A	35	5	40	87.5	12.5
Class B	28	12	40	70.0	30.0

Final result

- **Class A:** 35 Pass, 5 Fail → **87.5% Pass**
- **Class B:** 28 Pass, 12 Fail → **70% Pass**

These five codes are directly based on the datasets and tasks in your assigned problem bank.